

Social Agentics

Matt Ratto
Faculty of Information
University of Toronto
Toronto, Ontario, Canada
matt.ratto@utoronto.ca

Anastasia Kuzminykh
Faculty of Information
University of Toronto
Toronto, Ontario, Canada
anastasia.kuzminykh@utoronto.ca

Shion Guha
Faculty of Information
University of Toronto
Toronto, Ontario, Canada
shion.guha@utoronto.ca

Edith Law
School of Computer Science
University of Waterloo
Waterloo, Ontario, Canada
edithlaw@uwaterloo.ca

John Vines
Design Informatics
University of Edinburgh
Edinburgh, Scotland Uk
john.vines@ed.ac.uk

Abstract

Agentic AI is being heralded as the next step in the development of AI systems. Agentics, complex ensembles of different machine learning, data processing, and generative AI models, can provide new autonomous and proactive decision-making capabilities to organizations, participate in complex workflows, and, when needed, seek guidance from and provide insights to human users in natural languages. Collectively, we wish to explore how and why to design agentic systems to be situated within specific social and organizational contexts, the value of social theory and perspectives to this work, and the potential of this move to address critical issues with AI. Given the focus on social and organization context as essential to agentic design, we see this work as directly related to the CSCW community. We seek to bring together scholars from the diversity of disciplines within CSCW to develop research agendas, projects, and joint teaching initiatives that support the development of social agentic design and analysis.

CCS Concepts

• **Human-centered computing**; • **Collaborative and social computing**; • **Applied computing**; • **Law, social and behavioral sciences**; • **Computing methodologies**; • **Artificial intelligence**;

Keywords

Agentic AI, social theory, interdisciplinary approaches

ACM Reference Format:

Matt Ratto, Anastasia Kuzminykh, Shion Guha, Edith Law, and John Vines. 2025. Social Agentics. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3715070.3748276>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CSCW Companion '25, Bergen, Norway
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1480-1/2025/10
<https://doi.org/10.1145/3715070.3748276>

1 Introduction

This work starts from broad agreement that power relations, social structures, and social norms influence how human teams work and develop knowledge together. Our hypothesis is that better understanding these influences and incorporating explicit knowledge of them into agentic AI will result in systems more able to interoperate within diverse socio-cultural contexts and organizations, potentially addressing recognized issues with bias and equity. While a diversity of work similarly recognizes the need and value of social knowledge for agentic AI, some of the assumptions of this existing work require greater attention and validation. These include the assumption that since modern LLMs have been trained on human cultural context, they already contain an adequate understanding of human psychology, behaviors, and therefore social relations [1, 2]. We see a need to interrogate these assumptions and to explore alternative and additional starting places.

The overall goal of the SIG is to begin to constitute a shared area of practice from which research collaborations and shared projects can be developed. While it is our intention to cast our net widely in exploring the questions noted above, we see six existing areas of interdisciplinary work as providing useful insights to assist us in operationalizing social theory for agent development. For each area, we have also defined an initial question to help guide engagement.

1.1 Social Cognitive Architectures

We highlight the ongoing development of complex multi-agent systems by AI developers, based on social design patterns. Key examples include the LLM-as-judge frameworks [2, 2, 3] and multi-agent systems designed to engage in discussion, debate, deliberation, and argumentation [3–5]. These systems model human socio-cognitive processes but with little attention to the deep humanities and social science literature on these topics. How might socio-cognitive theories focused on distributed and collective decision-making be used to develop alternative AI architectures that support collectivist and constructivist forms of intelligence?

1.2 Social Simulation and Modeling

A deep literature exists, mostly predating new agentic technologies, that examines how computational agents can be used to explore social life [6–9]. Our goal is to explore how these techniques might

be used by agentic systems to better understand social complexity. How might agentic systems utilize techniques of social simulation to model and predict social interactions and outcomes, thereby improving agents' abilities to recognize, interpret, and respond to cultural and contextual differences.

1.3 Social Relations of AI

Existing literature in the humanities and social sciences addresses key concerns regarding AI [10–12], including critical literature exploring data, privacy, ethics, and labour [13–17]. While we intend our contributions to focus on design-level engagements and model systems, we see this literature as providing essential context and challenges we wish to address. What are the impacts of agentic systems within social structures, their potential and real impacts on equity, bias, and power within organizations, and how to resolve issues and create capacity through the development of new architectures and design practices?

1.4 Human-AI Teaming

We intend this work to supplement existing research on 'human-AI teaming' or HATs [18, 19], the study of AI and human organizations that mainly operates within technical fields. Some of the challenges raised within this literature are common to the exploration of AI in general, including questions of AI explainability and transparency, trust, and value alignment. But the HATs literature also highlights unique challenges that occur when developing agentic AI to support groups of humans working together. These include issues related to coordination and shared cognition [18] the need for systems to support the social development of human teams [20], and the capacity of such systems to develop 'situation awareness' [21]. Despite long-standing attention to human-AI teaming and its general and specific issues, we also note a lack of attention to social theory. How can enhanced engagement with social theory improve both the design and operation of human-AI teams?

1.5 Third-wave HCI

As generations of human-computer-interaction (HCI) practitioners have revealed, moving from individualistic accounts of human-computer interactions to accounts which center collaboration requires a concomitant move to incorporate understandings of human social and cultural dynamics into design and analysis. We see this move as particularly important to address in agentic AI, given its novel capacities to transform the relations between computational systems and human users. How can '3rd wave' HCI perspectives [22] better support methods for incorporating socio-cultural context into agentic AI?

1.6 Socially-situated AI

Scholars have noted how AI is socially-situated and begun to describe the need for methods and perspectives that take this into account. Examples of this work include research on the socially-situated nature of explainable AI [23] which leverages a method of Social Transparency (ST) to reveal and overcome the algorithm-centricity of most approaches. Experimental AI [24] notes the need to better understand the complex relations between humans and AI and notes the capacity of artist-led research to make visible the

social contexts in which human-AI interactions take place. Other researchers have focused on the need to situate AI systems within social contexts in order for these systems to better learn new concepts and to improve their interactions with people [25]. How can concepts of 'situatedness' help develop agentic AI?

1.7 Human-centricity and alternative approaches

Researchers have addressed how human-centricity can create bias and problematic environmental outcomes, defining 'posthuman' and 'more-than-human' approaches [26–36] as extending our understanding and focus beyond human-centred approaches. Applied to AI, these perspectives offer new design paradigms and patterns of making. How might we extend beyond the human in considering concepts of the social as applied to agentic AI?

2 Proposed Structure and Format

Our goal in developing this special interest group is to bring a diverse set of scholars together to pursue a shared agenda of incorporating social knowledge and perspectives into the design, operation, and analysis of agentic AI. Using the above topics, we will encourage participants to organize into smaller, focused groups and to map relevant concepts, methods, and related literature. To maximize accessibility, multiple formats for participation will be included; an online whiteboard; posterboard and post-it notes, and small and large-group discussions. Additionally, participants will be given a link to a discord server for follow-on engagements.

3 Biographical Sketches

Dr. Matt Ratto is the Associate Dean, Research and Professor, Faculty of Information, University of Toronto, Canada. Dr. Ratto's research explores the social production of knowledge in the context of emerging digital technologies. He leverages critical theories and perspectives from the humanities and social sciences to develop novel insights and design paradigms. He uses participatory design methods to collaborate with practitioners and is currently focused on the development of agentic AI systems to support human-AI teaming. Ratto will direct this project and act as lead coordinator.

Dr. Anastasia Kuzminykh is an Assistant Professor, Faculty of Information, University of Toronto, Canada. She explores the role of technology in Communication, Organization of Knowledge, and Information Ecosystems, focused on user's information sharing patterns and expectations of data dynamics with diverse types of systems. Most recently, her work addresses human-AI and human-human mediated communication. Dr. Kuzminykh will bring her expertise in psychology and AI to bear, supporting the theoretical work on socio-cultural cognitive theories.

Dr. Edith Law is an Associate Professor, school of computer science, University of Waterloo, Canada. She is broadly interested in social computing technology that coordinates small groups to large crowds, new models of interactions with machine intelligence, and how technology can be designed to foster and celebrate certain human values.

Dr. Shion Guha is an Assistant Professor, Faculty of Information, University of Toronto, Canada. His research interests include human-computer interaction, data science, and public policy. His

research has been key to the development of the field of Human-Centred Data Science. This intersectional research area combines technical methodologies with interpretive inquiry to address biases and structural inequalities in socio-technical systems.

Prof John Vines is Chair of Design Informatics in the School of Informatics at the University of Edinburgh, and co-directs the Institute for Design Informatics. His research focuses on how people experience and use digital and data-driven technologies in their everyday lives. He co-directs the Centre for Investing Innovation and the UKFin+ Network Plus.

4 Technical Requirements

We require a meeting room with a screen capable of viewing our preparatory slides and digital whiteboard. Multiple tables and chairs are preferred but not necessary.

5 Supplementary Materials

5.1 Community

We expect the SIG to draw upon CSCW scholars across a diversity of areas including but not limited to those working on Social Cognitive Architectures, Social Simulation and Modeling, Social Relations of AI, Human-AI Teaming, Third-wave HCI, Socially-situated AI, and critiques of Human-centricity. For each of these areas, agentic AI offers novel opportunities to extend upon past research and to connect with adjacent areas of practice. The planned SIG offers a venue for exploring these intersections.

5.2 Assumed attendee background

We hope to draw researchers from a wide variety of backgrounds, disciplinary homes, and domains/sectors of focused work. In our dissemination of a successful SIG proposal, we will encourage participation from graduate students, junior and senior researchers, and from different world regions.

5.3 The approach for organizing and presenting the SIG

Using the above topics, we will encourage participants to organize into smaller, focused groups and to map relevant concepts, methods, and related literature. To maximize accessibility, multiple formats for participation will be included; an online whiteboard; poster-board and post-it notes, and small and large-group discussions. Additionally, participants will be given a link to a discord server for follow-on engagements.

5.4 An informal schedule of discussion topics

Social Cognitive Architectures, Social Simulation and Modeling, Social Relations of AI, Human-AI Teaming, Third-wave HCI, Socially-situated AI, and critiques of Human-centricity.

5.5 A plan to attract a reasonable number of attendees

We will disseminate a successful SIG call broadly, leveraging our personal networks and access to email lists and digital platforms, including in CSCW-related fields such as STS, human-centred data science, informatics, design, and HCI.

5.6 Full name and contact information for the Organizer serving as the primary contact

Matt Ratto, matt.ratto@utoronto.ca

References

- [1] Alexander Sasha Vezhnevets, John P. Agapiou, Avia Aharon, Ron Ziv, Jayd Matyas, Edgar A. Duéñez-Guzmán, William A. Cunningham, Simon Osindero, Danny Karmon, and Joel Z. Leibo. 2023. Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia. <https://doi.org/10.48550/arXiv.2312.03664>
- [2] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua Gubler, Christopher Rytting, and David Wingate. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *Polit. Anal.* 31, 3 (July 2023), 337–351. <https://doi.org/10.1017/pan.2023.2>
- [3] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *CHI 2019*, May 2019. ACM. Retrieved from <https://www.microsoft.com/en-us/research/publication/guidelines-for-human-ai-interaction/>
- [4] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. <https://doi.org/10.48550/arXiv.2306.05685>
- [5] Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Saizhuo Wang, Kun Zhang, Yuanzhuo Wang, Wen Gao, Lionel Ni, and Jian Guo. 2025. A Survey on LLM-as-a-Judge. <https://doi.org/10.48550/arXiv.2411.15594>
- [6] Haitao Li, Qian Dong, Junjie Chen, Huixue Su, Yujia Zhou, Qingyao Ai, Ziyi Ye, and Yiqun Liu. 2024. LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods. <https://doi.org/10.48550/arXiv.2412.05579>
- [7] Mahmood Hegazy. 2025. Diversity of Thought Elicits Stronger Reasoning Capabilities in Multi-Agent Debate Frameworks. <https://doi.org/10.48550/arXiv.2410.12853>
- [8] Santi Ontañón and Enric Plaza. 2008. An Argumentation-Based Framework for Deliberation in Multi-agent Systems. In *Argumentation in Multi-Agent Systems*, Iyad Rahwan, Simon Parsons and Chris Reed (eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 178–196. https://doi.org/10.1007/978-3-540-78915-4_12
- [9] Tongxuan Liu, Xingyu Wang, Weizhe Huang, Wenjiang Xu, Yuting Zeng, Lei Jiang, Hailong Yang, and Jing Li. 2024. GroupDebate: Enhancing the Efficiency of Multi-Agent Debate Using Group Discussion. <https://doi.org/10.48550/arXiv.2409.14051>
- [10] Álvaro Carrera and Carlos A. Iglesias. 2015. A systematic review of argumentation techniques for multi-agent systems research. *Artif Intell Rev* 44, 4 (December 2015), 509–535. <https://doi.org/10.1007/s10462-015-9435-9>
- [11] Rosaria Conte, Bruce Edmonds, Scott Moss, and R. Keith Sawyer. 2001. Sociology and social theory in agent based social simulation: A symposium. *Computational & Mathematical Organization Theory* 7, 3 (2001), 183–205.
- [12] Batya Friedman and Helen Nissenbaum. 1997. Software agents and user autonomy. In *Proceedings of the first international conference on Autonomous agents (AGENTS '97)*, 1997. ACM, New York, NY, USA, 466–469. <https://doi.org/10.1145/267658.267772>
- [13] Nigel Gilbert. 2004. Agent-based social simulation: dealing with complexity. *The Complex Systems Network of Excellence* 9, 25 (2004), 1–14.
- [14] Stacy C. Marsella, David V. Pynadath, and Stephen J. Read. 2004. PsychSim: Agent-based modeling of social interactions and influence. In *Sixth International Conference on Cognitive Modeling*, 2004. Psychology Press, 243–248.
- [15] Jess Hohenstein, Rene F. Kizilcec, Dominic DiFranzo, Zhila Aghajari, Hannah Mieczkowski, Karen Levy, Mor Naaman, Jeffrey Hancock, and Malte F. Jung. 2023. Artificial intelligence in communication impacts language and social relationships. *Scientific Reports* 13, 1 (2023), 5487.
- [16] Septiani Amanda Puteri, Yulia Saputri, and Yarni Kurniati. 2024. The impact of artificial intelligence (AI) technology on students' social relations. *BICC Proceedings* 2, (2024), 153–158.
- [17] Anne Zimmerman, Joel Janhonen, and Emily Beer. 2024. Human/AI relationships: challenges, downsides, and impacts on human/human relationships. *AI and Ethics* 4, 4 (2024), 1555–1567.
- [18] Alessandro Mantelero. 2018. AI and Big Data: A blueprint for a human rights, social and ethical impact assessment. *Computer Law & Security Review* 34, 4 (2018), 754–772.
- [19] Moritz Altenried. 2020. The platform as factory: Crowdwork and the hidden labour behind artificial intelligence. *Capital & Class* 44, 2 (June 2020), 145–158. <https://doi.org/10.1177/0309816819899410>
- [20] Benedetta Catanzariti, Sravya Chandhiramowuli, Suha Mohamed, Sarayu Natarajan, Shantanu Prabhat, Noopur Raval, Alex S. Taylor, and Ding Wang.

2021. The Global Labours of AI and Data Intensive Systems. In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing*, October 23, 2021. ACM, Virtual Event USA, 319–322. <https://doi.org/10.1145/3462204.3481725>
- [19] Evgeni Aizenberg and Jeroen Van Den Hoven. 2020. Designing for human rights in AI. *Big Data & Society* 7, 2 (July 2020), 2053951720949566. <https://doi.org/10.1177/2053951720949566>
- [20] Sophie Berretta, Alina Tausch, Greta Ontrup, Björn Gilles, Corinna Peifer, and Annette Kluge. 2023. Defining human-AI teaming the human-centered way: a scoping review and network analysis. *Front. Artif. Intell.* 6, (September 2023). <https://doi.org/10.3389/frai.2023.1250725>
- [21] Jan B. Schmutz, Neal Outland, Sophie Kerstan, Eleni Georganta, and Anna-Sophie Ulfert. 2024. AI-teaming: Redefining collaboration in the digital era. *Current Opinion in Psychology* 58, (August 2024), 101837. <https://doi.org/10.1016/j.copsyc.2024.101837>
- [22] Katia Sycara and Michael Lewis. 2004. Integrating intelligent agents into human teams. In *Team cognition: Understanding the factors that drive process and performance*. American Psychological Association, Washington, DC, US, 203–231. <https://doi.org/10.1037/10690-010>
- [23] Kristin E. Schaefer, Edward R. Straub, Jessie Y. C. Chen, Joe Putney, and A. W. Evans. 2017. Communicating intent to develop shared situation awareness and engender trust in human-agent teams. *Cognitive Systems Research* 46, (December 2017), 26–39. <https://doi.org/10.1016/j.cogsys.2017.02.002>
- [24] Susanne Bodker. 2006. When second wave HCI meets third wave challenges. In *Proceedings of the 4th Nordic conference on Human-computer interaction: changing roles (NordiCHI '06)*, October 14, 2006. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/1182475.1182476>
- [25] Upol Ehsan, Q. Vera Liao, Michael Muller, Mark O. Riedl, and Justin D. Weisz. 2021. Expanding Explainability: Towards Social Transparency in AI systems. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*, May 07, 2021. Association for Computing Machinery, New York, NY, USA, 1–19. <https://doi.org/10.1145/3411764.3445188>
- [26] Drew Hemment, Ruth Aylett, Vaishak Belle, Dave Murray-Rust, Ewa Luger, Jane Hillston, Michael Rovatsos, and Frank Broz. 2019. Experiential AI. *AI Matters* 5, 1 (April 2019), 25–31. <https://doi.org/10.1145/3320254.3320264>
- [27] Ranjay Krishna, Donsuk Lee, Li Fei-Fei, and Michael S. Bernstein. 2022. Socially situated artificial intelligence enables learning from human interaction. *Proceedings of the National Academy of Sciences* 119, 39 (September 2022), e2115730119. <https://doi.org/10.1073/pnas.2115730119>
- [28] Laura Forlano. 2017. Posthumanism and Design. *She Ji: The Journal of Design, Economics, and Innovation* 3, 1 (March 2017), 16–29. <https://doi.org/10.1016/j.sheji.2017.08.001>
- [29] Christel Stalpaert, Kristof van Baarle, and Laura Karreman. 2021. *Performance and Posthumanism: Staging Prototypes of Composite Bodies* (1st ed.). Springer International Publishing AG, Cham. <https://doi.org/10.1007/978-3-030-74745-9>
- [30] Aykut Coskun, Nazli Cila, Iohanna Nicenboim, Christopher Frauenberger, Ron Wakkary, Marc Hassenzahl, Clara Mancini, Elisa Giaccardi, and Laura Forlano. 2022. More-than-human Concepts, Methodologies, and Practices in HCI. *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (April 2022), 1–5. <https://doi.org/10.1145/3491101.3516503>
- [31] Sarah Homewood, Marika Hedemyr, Maja Fagerberg Ranten, and Susan Kozel. 2021. Tracing Conceptions of the Body in HCI: From User to More-Than-Human. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (May 2021), 1–12. <https://doi.org/10.1145/3411764.3445656>
- [32] Ronald Wakkary. 2021. *Things we could design: for more than human-centered worlds*. The MIT Press, Cambridge, Massachusetts.
- [33] Rajakishore Nath and Riya Manna. 2023. From posthumanism to ethics of artificial intelligence. *AI & Soc* 38, 1 (February 2023), 185–196. <https://doi.org/10.1007/s00146-021-01274-1>
- [34] Charlotte Nordmoen. 2020. Decentring the Human in Digital Making - Towards Embodied Mattering. *Companion Publication of the 2020 ACM Designing Interactive Systems Conference* (July 2020), 543–547. <https://doi.org/10.1145/3393914.3395831>
- [35] Nandita Biswas Mellamphy. Re-thinking “Human-centric” AI: An Introduction to Posthumanist Critique. Retrieved August 8, 2022 from <https://www.europenowjournal.org/2021/11/07/re-thinking-human-centric-ai-an-introduction-to-posthumanist-critique/>
- [36] Matt Ratto, Sara Gram, Olivia Doggett, Peter Selby, Nadia Minian, Marta Maslej, Osnat Melamed, Jonathan Rose. Posthumanist AI Rethinking ‘the human’ as a model for generative AI. Workshop paper in GenAICHI: CHI 2024 Workshop on Generative AI and HCI.